

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 10

Tap periods (frames): [152, 178, 168, 155, 164, 185, 191, 167, 177]

Tap amplitudes (pixels): [198.2952344359289, 192.21342304844373, 206.1892334725555, 199.2636444512646, 191.02355875650522, 195.54283418218117, 204.9243762952568, 202.70421801235415, 186.34376834227646, 196.69773765857096]



Based on the extracted features provided, here is the analysis according to the MDS-UPDRS Item 3.4 criteria:

1. Amplitude (Decrement): The subject's tap amplitudes are highly consistent, ranging primarily between 190 and 200 pixels. The trend is stable (slope ≈ -0.3 pixels/tap), showing **no significant decrement** in amplitude throughout the 10-tap sequence. This rules out the decrement criteria found in scores 1, 2, 3, and 4.

2. Rhythm (Interruptions/Hesitations): The coefficient of variation for the tap periods is low ($\approx 7\%$), indicating a generally regular rhythm. There are no extreme outliers in the tap periods that would indicate a "halt" or "freeze" (longer arrest), nor are there 3-5 distinct interruptions.

3. Speed (Slowing): Analysis of the tap periods shows a trend of **slight slowing** (fatigue/sequence effect).

- **Average Period (First 4 taps):** ≈ 163 frames
- **Average Period (Last 4 taps):** ≈ 180 frames
- **Trend:** There is a positive slope in the periods ($\approx +2.4$ frames per tap), indicating the subject is getting slower as the task progresses.

Conclusion: While the subject maintains amplitude well and does not exhibit interruptions, the objective increase in tap period (slowing) fits the criterion for "Slight slowing". According to the scoring guidelines, "Any of the following: ...
b) Slight slowing" warrants a score of 1.

Score: 1